

Arbitrarily long runs of consecutive integers with equal Collatz height

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Abstract

For every prescribed length and lower bound, we construct a larger interval of consecutive positive integers that first reach one after the same number of shortcut Collatz steps, with equal odd-step counts. More generally, every finite pattern of nonnegative offsets has arbitrarily large translates with these properties. The proof first coalesces the pattern on a dyadic progression, then specializes its common endpoint to a power of two. Equal first hitting times also follow for the unshortened map. The principal statements and their supporting lemmas are formalized in Lean 4. These results do not establish convergence from an arbitrary prescribed starting integer.

1 Heights and the main result

Let $\mathbb{N}$ include zero. For positive integers, put

$$T(x) = \begin{cases} x/2, & x \text{ even,} \\ (3x+1)/2, & x \text{ odd.} \end{cases}$$

We use *height* for the first hitting time $h(x) = \min\{k \in \mathbb{N} : T^k(x) = 1\}$, taking $h(x) = \infty$ when that set is empty. In particular, $h(1) = 0$. Write $o_k(x)$ for the number of odd inputs among $x, T(x), \dots, T^{k-1}(x)$, with $o_0(x) = 0$. This is the height convention used by Garner [1], as distinguished from other conventions in LaDue [4, p. 156].

Garner’s 1985 run-length conjecture [1], as recorded by Lagarias [5, entry 72, p. 29], asserts that runs of consecutive integers with equal height can be arbitrarily long. Theorem 1 proves this assertion with equal odd counts as well, and with the runs occurring above every prescribed lower bound. A separate conjecture of Garner concerned a proposed characterization of equal-height consecutive pairs; the counterexamples of Elia and Tucker [6] concern that characterization, not the run-length assertion. Related studies include Gao’s computations of long runs [2] and LaDue’s results on clusters [4]. We give a self-contained construction and its Lean formalization, without claiming historical priority.

Theorem 1. *For every integer $\ell \geq 1$ and every $B \in \mathbb{N}$, there are an integer $x > \max\{1, B\}$ and $H, R \in \mathbb{N}$ such that*

$$h(x+i) = H, \quad o_H(x+i) = R \quad (0 \leq i < \ell).$$

Thus arbitrarily long runs of consecutive integers have the same finite Collatz height, and each prescribed run length occurs above every lower bound.

For example, 28, 29, 30 all reach 20 after seven steps and first reach one after thirteen steps, with five odd inputs each. We prove a stronger finite-pattern statement: any finite set $F \subseteq \mathbb{N}$ may replace $\{0, \dots, \ell-1\}$ in the theorem. The translating integer x is chosen by the construction; no convergence assertion for an arbitrary fixed x is assumed or deduced. The power-of-two endpoint argument uses the fact that two generates the units modulo powers of three, also used by Monks [3, Lemma 3.4] in back tracing.

2 Coalescence on a dyadic progression

We first record the dependence of a finite trajectory on its starting residue class.

Lemma 2. *For $c \in \mathbb{Z}_{>0}$, $k, t \in \mathbb{N}$, and $0 \leq j \leq k$,*

$$T^j(c + 2^k t) = T^j(c) + 2^{k-j} 3^{o_j(c)} t, \quad (1)$$

$$o_j(c + 2^k t) = o_j(c). \quad (2)$$

Proof. The assertions hold at $j = 0$. If $j < k$, the difference in (1) is even, so the two current integers have the same parity. An even step divides the difference by two; an odd step multiplies it by $3/2$ and adds one to both odd counts. This proves the induction. $\square$

In particular, the first k parities are constant on each positive residue class modulo 2^k . If a current value is $A + 3^r t$, either parity can be prescribed by restricting t to $\varepsilon + 2u$, where $\varepsilon \in \{0, 1\}$. After the prescribed actual step, the coefficient of u is 3^r or 3^{r+1} according as that step is even or odd. Thus this prescription can be repeated, and each additional step doubles the parameter modulus.

Lemma 3. *For every $n \in \mathbb{N}$, there are $k \in \mathbb{N}$ and $a \in \mathbb{Z}_{>0}$ such that, for every $t \in \mathbb{N}$,*

$$T^k(a + 2^k t) = T^k(a + 2^k t + n),$$

$$o_k(a + 2^k t) = o_k(a + 2^k t + n).$$

Proof. We use strong induction on n . For $n = 0$, take $k = 0$ and $a = 1$. For $n = 1$, the two trajectories

$$4 + 8t \mapsto 2 + 4t \mapsto 1 + 2t \mapsto 2 + 3t,$$

$$5 + 8t \mapsto 8 + 12t \mapsto 4 + 6t \mapsto 2 + 3t$$

each have one odd step. This gives $k = 3$ and $a = 4$.

If $n = 2m > 0$, use the progression $a + 2^k t$ for m supplied by induction. The starts $2a + 2^{k+1}t$ and $2a + 2^{k+1}t + 2m$ take one even step to that pair. They therefore coalesce after $k + 1$ steps, with equal odd counts.

It remains to treat odd $n > 1$. Choose m, c, d by

$n \bmod 4$	m	c	d
1	$(3n + 1)/4$	1	2
3	$(3n - 1)/4$	2	1

so that $0 < m < n$ and $3n + d = 4m + c$. Let $a + 2^k t$ be the progression for m . Because 2^k is invertible modulo three, some $q \in \{0, 1, 2\}$ makes $4(a + 2^k q) - c$ divisible by three. The integer

$$a' = \frac{4(a + 2^k q) - c}{3}$$

is positive. For $X = a' + 2^{k+2}t$ and $Y = a + 2^k(q + 3t)$, we have

$$3X + c = 4Y, \quad 3(X + n) + d = 4(Y + m).$$

If $3z + 1 = 4w$, then $z \equiv 1 \pmod{4}$ and its next two steps are odd, even, with endpoint w . If $3z + 2 = 4w$, then $z \equiv 2 \pmod{4}$ and its next two steps are even, odd, again with endpoint w . Hence X and $X + n$ reach Y and $Y + m$ after two steps, each having used one odd step. The induction hypothesis at parameter $q + 3t$ finishes the construction. $\square$

2.1 Synchronizing affine families

To add an offset to a collection that already coalesces, we must synchronize its current trajectory with the common trajectory of the collection. Their coefficients in the remaining parameter can be different powers of three.

Lemma 4. *Let $A, B \in \mathbb{Z}_{>0}$ and $r, s \in \mathbb{N}$. There exist $\ell, t_0, Q \in \mathbb{N}$ and $C \in \mathbb{Z}_{>0}$ such that, for every $v \in \mathbb{N}$,*

$$T^\ell(A + 3^r(t_0 + 2^\ell v)) = T^\ell(B + 3^s(t_0 + 2^\ell v)) = C + 3^Q v. \quad (3)$$

The two length- ℓ trajectories have fixed odd counts u, w satisfying $r + u = s + w = Q$.

Proof. Exchange the families if necessary so that $r \leq s$. Initially their current values satisfy

$$Y = 3^d X + D, \quad d = s - r, \quad D = B - 3^d A \in \mathbb{Z}.$$

At every subsequent stage the coefficient of the remaining parameter in X is a power of three and is therefore odd. We may prescribe its next parity, as above. The displayed relation determines the parity of Y . When $d > 0$, the following steps reduce the relation to one with a smaller exponent.

If $D \neq 0$ is even, prescribe X even. Both values take an even step, and

$$Y' = 3^d X' + D/2.$$

Repeating this finitely many times makes D odd. This also applies to negative D , by dividing out the power of two in $|D|$. If $D = 0$, prescribe X odd. Both values take an odd step, and

$$Y' = 3^d X' + \frac{1 - 3^d}{2}.$$

The new constant is nonzero because $d > 0$, so the preceding halving procedure makes it odd. Once D is odd, prescribe X odd. Then Y is even, and

$$X' = \frac{3X + 1}{2}, \quad Y' = \frac{Y}{2}, \quad Y' = 3^{d-1} X' + \frac{D - 3^{d-1}}{2}. \quad (4)$$

The last constant is an integer and the exponent has decreased. There are finitely many steps at each fixed positive d and finitely many decreases of d , so eventually $d = 0$.

The current values now have the same coefficient in the remaining parameter and differ by a constant integer D . If $D = 0$, they are equal already. Otherwise, apply Lemma 3 with gap $|D|$ to the smaller current value. Suppose its required progression is $a + 2^k z$. The smaller value has the form $E + 3^q u$, so the congruence

$$E + 3^q u_0 \equiv a \pmod{2^k}$$

has a solution. Choose a nonnegative representative u_0 large enough that $E + 3^q u_0 \geq a$. On the parameter class $u = u_0 + 2^k v$, the smaller value belongs to the positive tail of the required progression for every $v \geq 0$. The next k steps therefore make the two values equal.

Each parameter restriction has modulus two for each added actual step. Their composition has the form $t = t_0 + 2^\ell v$, where ℓ is the common total number of steps. All bases and current values remain positive. By Lemma 2, the resulting odd counts u, w are independent of v , and the two final affine coefficients are 3^{r+u} and 3^{s+w} . Equality at $v = 0$ and $v = 1$ forces these coefficients to agree. This proves (3) and $r + u = s + w = Q$. $\square$

Proposition 5. *For every finite $F \subset \mathbb{N}$, there are $K, R \in \mathbb{N}$ and $a, b \in \mathbb{Z}_{>0}$ such that, for every $n \in F$ and $t \in \mathbb{N}$,*

$$T^K(a + 2^K t + n) = b + 3^R t, \quad o_K(a + 2^K t + n) = R. \quad (5)$$

Proof. Enlarge F to include zero, and enumerate its offsets starting with zero. For the initial singleton take $K = R = 0$ and $a = b = 1$. Suppose the assertion holds for the offsets already included. Their common current value on $a + 2^K t$ is $U(t) = b + 3^R t$. For the next offset n , Lemma 2 gives

$$T^K(a + 2^K t + n) = d + 3^S t, \quad d = T^K(a + n) > 0, \quad S = o_K(a + n).$$

Apply Lemma 4 to these two affine families. On a class $t = t_0 + 2^\ell v$, they have the same image after ℓ additional steps. Every previously included trajectory starts those steps at exactly $U(t)$, so all previous equalities are preserved.

The new starting progression is

$$(a + 2^K t_0) + 2^{K+\ell} v.$$

Its modulus agrees with the new common time $K + \ell$. If the two added trajectories use u, w odd steps, the last assertion of Lemma 4 gives $R + u = S + w$. Thus all total odd counts agree as well. The new endpoint has the required form $b' + 3^{R+u} v$. Finite induction proves the proposition. $\square$

3 Equal heights by endpoint specialization

Lemma 6. *Let $b \in \mathbb{Z}_{>0}$ with $3 \nmid b$, and let $R \in \mathbb{N}$. There are arbitrarily large $t \in \mathbb{N}$ for which $b + 3^R t$ is a power of two.*

Proof. We give the residue argument explicitly. For $m \geq 1$,

$$4^{3^{m-1}} \equiv 1 + 3^m \pmod{3^{m+1}}. \quad (6)$$

The case $m = 1$ is immediate. Cubing a number of the form $1 + 3^m(1 + 3c)$ proves the next case, since the terms beyond the linear term are divisible by 3^{m+2} .

Powers of two represent both units modulo three. Suppose $2^s \equiv b \pmod{3^m}$. Multiplying 2^s by $(4^{3^{m-1}})^j$, for $j = 0, 1, 2$, changes its residue modulo 3^{m+1} by $j2^s 3^m$, by (6). Since $3 \nmid 2^s$, these are the three possible lifts of its residue modulo 3^m . One therefore represents b modulo 3^{m+1} . Induction gives $2^s \equiv b \pmod{3^R}$ for every $R \geq 1$; the case $R = 0$ is immediate.

The same identity gives $2^{2 \cdot 3^R} \equiv 1 \pmod{3^R}$. We may consequently increase s by arbitrary multiples of $2 \cdot 3^R$ while preserving the residue. For all sufficiently large such exponents,

$$t = \frac{2^s - b}{3^R}$$

is a nonnegative integer, and these values of t are unbounded. $\square$

Proof of Theorem 1. We prove the stronger assertion for any finite set $F \subseteq \mathbb{N}$ of offsets, which includes $F = \{0, \dots, \ell - 1\}$. Apply Proposition 5 to $F \cup \{0, 1\}$, obtaining K, R, a, b . We first show $3 \nmid b$. An odd step has output congruent to two modulo three. Thus a current value divisible by three has only an even predecessor, which is also divisible by three. If $3 \mid b$, every step in any length- K trajectory ending at b would be even, and its initial value would equal $2^K b$. The distinct starts a and $a + 1$ both end at b , a contradiction.

By Lemma 6, choose $t \geq 1$ as large as necessary, with $b + 3^R t = 2^s$, and put $x = a + 2^K t$. The common endpoint after K steps is 2^s . After s further even steps, all trajectories reach one, with total odd count R . Taking t large ensures $x > \max\{1, B\}$.

It remains to check that this is the first visit to one. For $j < K$, Lemma 2 gives

$$T^j(x + n) = T^j(a + n) + 2^{K-j} 3^{o_j(a+n)} t > 1.$$

For $j = K + q < K + s$, the value is $2^{s-q} > 1$. Thus $h(x + n) = H = K + s$ and $o_H(x + n) = R$ for every $n \in F$. Taking $F = \{0, \dots, \ell - 1\}$ gives the theorem. $\square$

For the unshortened map that sends odd x to $3x + 1$, each of the R odd steps above expands to an odd step followed by an even step. The corresponding first hitting time is therefore $H + R$, also common to the whole pattern. The inserted intermediate values are greater than one, so first arrival is preserved.

4 Formal verification

The independent artifact contains nine Lean modules, with 66 public theorems and lemmas and eight definitions. A fresh build and axiom audit checks all 74 declarations using Lean 4.27.0 and mathlib commit

`a3a10db0e9d66acbebf76c5e6a135066525ac900.`

Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. No proof placeholder, added mathematical axiom, or numerical search result is used as proof. The entry point is `formal/FinitePatternConvergence.lean`; its two theorems state the arbitrary finite-pattern result and the consecutive-run corollary. Both explicitly require every earlier iterate to exceed one. Appendix A reproduces the definitions and theorem types. The elementary conversion to the unshortened map is a prose consequence; the Lean endpoints use T as defined above.

The `equal-collatz-heights/` directory contains the source, PDF, claim map, pinned verifier, and retained build report. The repository is

`https://github.com/x1xh101/collatz-two-coordinate-obstruction`.

From that directory, run `python3 verify.py` with `--mathlib-root` pointing to the pinned check-out and `--report` to an unused output path. The accompanying README gives the complete setup. Kernel acceptance verifies the formal proposition; matching that proposition to the mathematical statement is a separate reading task addressed by the appendix and claim map.

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References

- [1] L. E. Garner. *On heights in the Collatz $3n + 1$ problem*. Discrete Mathematics **55**(1), 57–64, 1985. [https://doi.org/10.1016/S0012-365X\(85\)80020-0](https://doi.org/10.1016/S0012-365X(85)80020-0).
- [2] G.-G. Gao. *On consecutive numbers of the same height in the Collatz problem*. Discrete Mathematics **112**(1–3), 261–267, 1993. [https://doi.org/10.1016/0012-365X\(93\)90240-T](https://doi.org/10.1016/0012-365X(93)90240-T).
- [3] K. M. Monks. *The sufficiency of arithmetic progressions for the $3x + 1$ Conjecture*. Proceedings of the American Mathematical Society **134**(10), 2861–2872, 2006. <https://doi.org/10.1090/S0002-9939-06-08567-4>. Author version: <https://monks.scranton.edu/files/pubs/SufficiencyRev4.pdf>.
- [4] M. D. LaDue. *Clusters of integers with equal total stopping times in the $3x + 1$ problem*. The Fibonacci Quarterly **56**(2), 156–162, 2018. <https://www.fq.math.ca/Papers1/56-2/ladue121517.pdf>.
- [5] J. C. Lagarias. *The $3x + 1$ Problem: An Annotated Bibliography (1963–1999)*. arXiv:math/0309224v13, 2011, entry 72, p. 29. <https://arxiv.org/pdf/math/0309224v13>.
- [6] M. Elia and A. Tucker. *Consecutive Integers and the Collatz Conjecture*. arXiv:1511.09141, 2015. <https://arxiv.org/abs/1511.09141>.

A Exact Lean declarations

The following declarations use namespace `CollatzPositiveProgression`. Natural numbers include zero. The constructed starts are positive. Here `oddCount k x` counts odd inputs in the first k steps. The first-hitting-time theorem requires every earlier value to exceed one.

The map, its iterates, and its odd-step count.

`formal/PositiveProgressionCoalescence.lean`, lines 9–18.

```

9 def step (x : ℕ) : ℕ :=
10   if x % 2 = 0 then x / 2 else (3 * x + 1) / 2
11
12 def iterate : ℕ → ℕ → ℕ
13   | 0, x => x
14   | k + 1, x => iterate k (step x)
15
16 def oddCount : ℕ → ℕ → ℕ
17   | 0, _ => 0
18   | k + 1, x => x % 2 + oddCount k (step x)

```

Coalescence on a common dyadic progression.

`formal/FinitePatternCoalescence.lean`, lines 50–53.

```

50 theorem every_finite_pattern_coalesces (F : Finset ℕ) :
51   ∃ K R a b : ℕ, 0 < a ∧ 0 < b ∧ ∀ n ∈ F, ∀ t : ℕ,
52     iterate K (a + 2 ^ K * t + n) = b + 3 ^ R * t ∧
53     oddCount K (a + 2 ^ K * t + n) = R

```

Equal finite first hitting times for any finite pattern.

`formal/FinitePatternConvergence.lean`, lines 8–12.

```

8 theorem every_finite_pattern_has_equal_first_hitting_times
9   (F : Finset ℕ) (lower : ℕ) :
10   ∃ x H R : ℕ, lower < x ∧ 1 < x ∧ ∀ n ∈ F,
11     iterate H (x + n) = 1 ∧ oddCount H (x + n) = R ∧
12     ∀ j < H, 1 < iterate j (x + n)

```

Consecutive runs of every prescribed length.

`formal/FinitePatternConvergence.lean`, lines 32–36.

```

32 theorem arbitrarily_long_consecutive_equal_first_hitting_times
33   (length lower : ℕ) :
34   ∃ x H R : ℕ, lower < x ∧ 1 < x ∧ ∀ n < length,
35     iterate H (x + n) = 1 ∧ oddCount H (x + n) = R ∧
36     ∀ j < H, 1 < iterate j (x + n)

```